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PERFORMANCE SPECIFICATION

PROPELLANT, HYDRAZINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

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1. SCOPE

1.1 Scope. This specification covers the requirements for two grades of hydrazine (N_2H_4).

1.2 Classification. Hydrazine will be of the following designated grades as specified (see 6.2).

GRADES

Monopropellant (MPH)	-	Legacy product
High Purity (HPH)	-	Normal production product.

1.3 Part or identifying number (PIN). The PINs to be used for hydrazine acquired to this specification are created as follows:

M 26536 – X Example of PIN: M26536-H

- X: Grade (see 1.2). Use “M” for monopropellant; “H” for high purity.
- 26536: Specification number.
- M: Prefix for military specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25604	Propellant, uns-Dimethylhydrazine
MIL-PRF-27404	Propellant, Monomethylhydrazine

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM D1193	Standard Specification for Reagent Water
ASTM D1976	Standard Test Method for Elements in Water by Inductively-Coupled Plasma Atomic Emission Spectroscopy
ASTM D2276	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1042	Laboratory Glassware – One-Mark Volumetric Flasks

(Copies of these documents are available from https://www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical properties. The chemical and physical properties of the propellant shall conform to those listed in Table I when tested in accordance with (IAW) the applicable test methods.

3.2 Limiting values. The following guidelines apply to all specified limits within this specification. To determine conformance, round observed or calculated values to the nearest unit in the last right-hand digit of the specification limit, following the Rounding Method of ASTM E29.

3.3 Filter. A filter with a rating of at least 2 µm (micrometer) nominal and 10 µm absolute rating shall be installed between the manufacturer's plant system and the container to be filled for delivery.

3.4 Qualitative. The propellant shall appear as a colorless, homogeneous liquid upon visual examination under transmitted light.

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TABLE I. *Chemical and physical properties.*

Properties	Monopropellant Grade (MPH)	High Purity Grade (HPH)	Test Paragraph
Hydrazine wt%, min.	98.5	99.0	4.3.2
Water wt%, max.	1.0	0.75	4.3.2
Ammonia wt%, max.	---	0.3	4.3.2
Particulate mg/L, max.	1.0	1.0	4.3.3
Chloride mg/kg (ppmw), max.	5	5	4.3.4
Cadmium mg/kg (ppmw), max.	----	0.04	4.3.5
Iron mg/kg (ppmw), max.	20	1.0	4.3.5
Silicon mg/kg (ppmw), max.	----	0.05	4.3.5
Non-volatile residue mg/L, max.	50	10	4.3.6
Carbon dioxide(1) wt%, max.	0.003	0.003	4.3.7
Aniline wt%, max	0.50	0.003	4.3.8
Carbonaceous Material(2) ,wt%, max.	0.02	0.005	4.3.8
Toluene (ppmw), max.(3)	----	1	4.3.8

NOTES:

(1) Determination of carbon dioxide (CO_2) content shall be performed before all other analyses to prevent further absorption of CO_2.

(2) Excluding Aniline

(3) Toluene requirement applies at point of manufacture only.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspections.

4.2 Conformance inspection. The conformance inspection shall consist of the following:

a. Individual tests (see 4.2.1).

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b. Sampling tests (see 4.2.2).

4.2.1 Individual tests. The following test shall be conducted on the propellant, as described under 4.3:

a. Examination of product (see 4.3.1).

4.2.2 Sampling tests. The propellant shall be selected according to 4.2.2.1 and subjected to the tests in Table I as described under 4.3.

4.2.2.1 Sampling plan. Unless otherwise specified (6.2), each filled shipping container shall be considered a lot and shall be sampled.

4.2.2.1.1 Sample. A sample consists of 800 to 1000 milliliters (mL) of propellant for all tests specified in 4.2.2. Unless otherwise specified, conformance tests shall be made on the sample of propellant taken directly from the shipping container. When required, the sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the conformance tests specified herein. The clear glass narrow mouth bottles intended for sampling shall be specially cleaned and handled according to the procedure described in ASTM D2276 Section 10 (Preparation of Flushing Apparatus) with the exception listed in 4.3.3.d. Sampling shall be conducted in such a manner that atmospheric exposure of the contents of the shipping container and the sample is minimized.

4.2.3 Rejection. When any sample of the propellant tested IAW 4.3 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.3 Test methods.

4.3.1 Examination of product. The propellant shall be visually examined to ensure that the material conforms to 3.4. The examination shall be conducted after a representative aliquot of the sample has been transferred to a clear glass vessel and compared to the same type vessel containing the same volume of purified water.

4.3.2 Hydrazine assay, ammonia, and water. The hydrazine assay (N_2H_4), ammonia (NH_3) (HPH only), and water (H_2O) content of the propellant shall be determined by the gas chromatographic (GC) method described in Appendix A.

4.3.3 Particulate. The propellant sample shall be tested gravimetrically for particulate contamination in accordance with ASTM D2276, with the following exceptions:

a. Mix the sample thoroughly without exposure to air. Immediately pour no less than 250 mL of the sample into a clean graduated cylinder.

b. Pass the sample aliquot through a Mitex disc polytetrafluoroethylene (PTFE) hydrophobic, 5 µm, 47 mm, white filter or equivalent.

c. The oven temperature shall be set at 110 °C (230 °F).

d. Filtered American Chemical Society (ACS) reagent (or higher) grade isopropyl alcohol shall be used for rinsing the graduated cylinder and filter holder instead of petroleum ether specified in ASTM D2276. Use a suitable solvent resistant filter disc with an absolute rating of at least 1.0 µm. Wet the filter with the filtered isopropyl alcohol prior to filtering the sample.

4.3.4 Chloride. The chloride (Cl^-) content of the propellant shall be determined by the ion chromatography method described in Appendix B.

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4.3.5 Elemental analyses. The cadmium (Cd), iron (Fe), and silicon (Si) content of the propellant shall be determined by the inductively coupled plasma (ICP) method described in Appendix C.

4.3.6 Non-volatile residue. The non-volatile residue (NVR) of the propellant shall be determined in duplicate by one of the methods in Appendix D. In case of a dispute, the hot plate method (D.2.1) shall be the referee.

4.3.7 Carbon dioxide. The carbon dioxide (CO_2) content of the propellant shall be determined by the method described in Appendix E.

4.3.8 Carbonaceous material, aniline, and toluene. Carbonaceous material (excluding aniline), aniline, and toluene (at point of manufacture) shall be determined by the gas chromatographic method in Appendix F.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant described by this specification is intended for use as a fuel in both military and civil rocket engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade of the propellant required (1.2).

c. Method of shipment, type and capacity of containers.

d. Quantity by weight in pounds (avoirdupois).

e. When a different sampling plan is required (4.2.2.1).

f. When test methods or limits are other than those specified (4.3).

g. Packaging requirements (5.1).

6.3 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with the 29CFR1910.1200.

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6.4 Subject term (key word) listing.

Catalytic Engine

Fuel

Monopropellant

Rocket Engine

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

HYDRAZINE ASSAY, AMMONIA, AND WATER

A.1 SCOPE

A.1.1 Scope. This Appendix describes the gas chromatographic (GC) method for the determination of the hydrazine assay, ammonia, and water content in the sample. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 METHODS

A.2.1 Gas chromatographic method - capillary column.

A.2.1.1 Procedure. Use a 30- or 60-meter (m) × 0.53 millimeter (mm), 1.00 µm film thickness, wax-phase (DB-Wax, Stabilwax DB, or equivalent) column, a thermal conductivity detector (TCD) with a split injector, and an ultra-inert, deactivated, glass wool packed split liner. Table A-I summarizes the critical operating parameters. Note that these parameters may vary depending on the GC system and detector brands and may be adjusted by the chemist to optimize component resolution.

Allow the instrument to stabilize and then condition the column by making at least two injections of hydrazine (HPH or MPH, depending on sample) before the sample sequence. Perform analysis by processing three consecutive 1 microliter (µL) injections of the sample. The three analysis runs must meet the quality control criteria in A.2.1.3.

TABLE A-I. *Recommended GC parameters.*

PARAMETER	CONDITION[1]
Injector Port Temp	150 °C
Injection Mode	Split injection
Sample Volume.	1 µL
Split Ratio	10:1
Carrier Gas	He or H_2
Control Flow	10 mL/min
Control Mode	Flow
TCD Detector Temp.	210 °C
Equilibration Time	2.00 min
Initial Temp.(T_0)	60 °C
Initial Time (t_0)	3.00 min
Rate (1)	10 °C/min
Final Temp. (T_{f1})	100 °C
Final Time (t_{f1})	0.00 min
Rate (2)	20 °C/min
Final Temp. (T_{f2})	185 °C
Final Time (t_{f2})	3.75 min
Note 1 – Parameters identified are for a He carrier. H_2 carrier may require adjustments.	

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Elution order should be:
Ammonia > Water > Hydrazine

A.2.1.2 Calculations.

$$\% N_2H_4 = \frac{A_{N_2H_4} \times 100}{\sum A_i}$$

$$\% H_2O = \frac{A_{H_2O} \times 100}{\sum A_i}$$

$$\% NH_3 = \frac{A_{NH_3} \times 100}{\sum A_i}$$

NOTE: % Ammonia (NH_3) for high purity grade only.

where,

$A_{N_2H_4}$= The measured area of the N_2H_4 peak

A_{H_2O} = The measured area of the H_2O peak

A_{NH_3} = The measured area of the NH_3 peak

$\sum A_i$ = The sum of all measured peak areas

Assumption: The thermal conductivities of all components in the sample are equal.

A.2.1.3 Quality control. The following performance criteria must be met for fuel purity analysis:

GC replicate analyses (3 consecutive) on Hydrazine peak area	RSD[1] ≤ 5 %

1. RSD = Standard Deviation * 100% / Mean

A.2.1.4 Equipment.

(1) Gas chromatograph equipped with a TCD and a split injector.

(2) Data acquisition software (e.g. Agilent ChemStation or equivalent).

(3) A 30 or 60 m x 0.53 mm ID and 1.00 µm film thickness, wax phase column (DB-wax, Stabilwax DB, or equivalent).

(4) Ultra-inert, deactivated, 4 mm diameter, split, straight, glass wool liner.

(5) No-bleed septa.

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CHLORIDE

B.1 SCOPE

B.1.1 Scope. This appendix describes the methods for determining the chloride content of hydrazine. It is a mandatory part of the specification. In the case of a dispute, the ion chromatography shall be the referee method. The information herein is intended to ensure compliance.

B.2 METHODS

B.2.1 Ion chromatography (IC) method.

B.2.1.1 Procedure. To the hydrazine NVR sample in the designated vessel (refer to section 4.3.6), add sufficient purified water (18 M-ohm resistivity or greater) to achieve an instrument reading within the calibrated range. Gently warm the mixture for 3 to 5 minutes, ensuring the water contacts all interior surfaces of the vessel previously exposed to the fuel mixture. Avoid boiling the solution. Allow the solution to cool completely. Then, transfer a representative aliquot to a labeled IC vial for analysis.

B.2.1.2 Calibration. Prepare a series of at least three calibration standards in purified water to bracket the expected concentration range of the samples by combining appropriate aliquots of a certified Cl^- or multi-element IC anion standard solution. Prepare and/or commercially obtain a mid-range check standard from a different stock standard source to verify the calibration (Initial Calibration Verification, ICV). Prepare a calibration curve by plotting the instrument response (integrated peak area) vs. the Cl^- standard concentration to determine the sample concentration. Alternatively, use the quantitation method recommended by the instrument manufacturer.

B.2.1.3 Quality control.

(1) The calibration correlation coefficient must be ≥ 0.995.

(2) Analyze the ICV every 10 samples to validate the instrument calibration. The ICV should fall within ± 10% of its concentration.

B.2.1.4 Calculation. Calculate the chloride content of the hydrazine using the calibration curve generated by the instrument software.

B.2.1.5 Reagents.

a. Purified water of 18 M-ohm or greater resistivity.

b. Eluent mix stock preparation approximately equivalent to 3.5 mM sodium carbonate/1.0 mM sodium bicarbonate or 30 mM potassium hydroxide (depending on column and instrument configuration):

(1) For IC systems with an automatic eluent generator: Follow the instrument owner's manual instructions.

(2) For manual preparation:

If using commercially available concentrate: Combine an appropriate aliquot of a commercially available IC sodium carbonate/sodium bicarbonate concentrate in a volumetric flask containing purified water. Fill to the mark and mix thoroughly. If using in-house preparation: Dissolve approximately 371 mg of ACS reagent-grade sodium carbonate and 84 mg of sodium bicarbonate in a 1 L volumetric flask. Record the weights of the sodium carbonate and sodium bicarbonate to the nearest 1 mg. Fill to the mark with purified water and mix thoroughly.

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c. Chloride or multi-element IC anion standard solution (must be certified).

B.2.1.6 Equipment. The ion chromatograph must have the following components:

(1) Eluent pump capable of delivering 1 - 5 mL/min of eluent at a nominal flow and pressure specified by the instrument manufacturer.

(2) Sample pump capable of filling a sample loop with no resistance.

(3) Sample loop capable of delivering 5 to 100 µL of sample to the analytical column.

(4) An ion exchange column suitable for resolving chloride.

(5) A suppressor device.

(6) Conductivity cell.

(7) Data acquisition device.

(8) Five decimal place analytical balance (required only if eluent mix is prepared in-house).

B.2.2 Colorimetric method.

B.2.2.1 Procedure. Add to the propellant residue container in one vessel from 4.3.6 five milliliters of ferric alum indicator and transfer to a 100 mL volumetric flask. Rinse the vessel with three 5 mL purified water aliquots and combine into the volumetric flask. Set samples aside. Prepare a calibration blank and at a minimum two standards to bracket the expected chloride concentration in the samples in 100 mL volumetric flasks and add 5 mL of ferric alum indicator. Add 2.5 mL of mercuric thiocyanate solution to each volumetric flask, samples and standards, and dilute to the mark with purified water. Mix the solutions thoroughly and allow to stand in darkness for 15 – 30 min. Analyze in a UV-Vis spectrophotometer at 460 nm, at the wavelength recommended by the instrument manufacturer, or at that which provides the highest sensitivity using 5 cm cells. Use the blank to set the instrument to zero absorbance and determine the absorbances of standards and samples. Prepare a calibration curve of absorbance vs concentration and determine the chloride concentration in the samples. The calibration correlation coefficient shall be ≥ 0.999.

B.2.2.2 Calculation. Calculate the percent chloride in the sample using the following formula:

$$\%Cl^{-} = \frac{(ppmv\ Cl^{-})(V_d)}{(V_h)(D)(1x10^4)}$$

where,

ppmv Cl^- = Concentration of Cl^- determined from calibration curve

V_d = Final dilution volume taken from NVR residue in mL.

V_h = Volume of hydrazine used in NVR analysis in mL.

D = Density of hydrazine (1.01 g/mL).

B.2.2.3 Reagents and equipment.

B.2.2.3.1 Reagents.

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a. Purified water of 18M-ohm or greater resistivity.

b. Ferric alum indicator, Volhard (oxidation reduction indicator) – Dissolve 10 g of ferric ammonium sulfate dodecahydrate ($FeNH_4(SO_4)_2 \cdot 12H_2O$) ACS reagent grade in approximately 50 mL of purified water. Add 10 mL of ACS reagent grade nitric acid and dilute to 100 mL with purified water. (may be commercially obtained).

c. Mercuric thiocyanate, 4.17 g/L in methanol for chloride determination (commercially available).

d. Chloride standard, 1000 mg/L (may be commercially obtained; must be certified).

B.2.2.3.2 Equipment. UV-Vis Spectrophotometer, double beam, equipped with 5 cm cells.

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ELEMENTAL ANALYSIS

C.1 SCOPE

C.1.1 Scope. This Appendix describes the inductively coupled argon plasma (ICP) method for the determination of Cd, Fe, and Si content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 METHODS

C.2.1 Inductively coupled argon plasma. For each instrument, interference effects must be evaluated, corrected as applicable (e.g. ASTM D1976 for atomic emission spectroscopy (AES)), and instrument performance validated. The argon gas supply must have a purity of 99.99% or better. Choose the elements wavelengths recommended by the instrument manufacturer or that which provide the highest sensitivity. In the case of a dispute, ICP-AES shall be the referee method.

C.2.1.1 Procedure. Rinse the hydrazine NVR (if being used for D.2.1 NVR analysis) contained in one vessel from D.2.1 as follows: Add 20 mL of a 5% (v/v) solution of concentrated HNO_3 in purified water to the vessel and place on a hot plate. Heat the mixture to near boiling, maneuver vessels so the acid contacts the surface area of the vessel that the fuel mixture came in contact with and maintain at that temperature for three to five minutes. Allow the solution to cool, mix thoroughly by swirling to homogenize, and transfer to a 50 mL graduated polypropylene (PP) ICP vial/digestion cup. Adjust the volume back to 20 mL and analyze.

C.2.1.2 Calibration. Prepare an acid matrix blank and a multi-element standard containing the elements of interest in a 5% (v/v) solution of HNO_3 in purified water solution to bracket the expected concentration range of the samples by combining an appropriate aliquot of a certified multi-element stock standard in a Class A polymethylpentene (PMP) or perfluoroalkoxy- copolymer volumetric flask. Prepare a mid-range check standard, from a different stock standard source to verify the calibration. Determine the concentration from the calibration curve prepared from the multi-element standard using the acid matrix blank as the zero point, and record the Cd, Fe, and Si concentrations of the sample.

C.2.1.3 Quality control. The instrumental drift shall be checked using a combination of the acid matrix blank, internal standard, and multi-element standard at the beginning of each run, at least every ten readings during the run, and at the end of the run. The blank shall read ≤ the detection limit and the multi-element standard shall fall within ± 10% of the standard value for each of the elements of interest. If the recovery of the analyte falls outside of the required control limits, the source of the problem must be identified and resolved before continuing analyses.

C.2.1.4 Calculation. Calculate the cadmium, iron, and silicon content of the hydrazine using the following formula:

$$Analyte\ ppmw = \frac{\left(Analyte_{ppmv}\right)(V_d)}{(V_p)(\rho)}$$

where,

$Analyte_{ppmv}$ = Analyte content of the sample solution from the calibration curve.

V_d = Final dilution volume taken from NVR in mL.

V_p = Volume of hydrazine used in the NVR analysis in mL.

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ρ = Density of hydrazine (1.01 g/mL).

C.2.1.5 Reagents and equipment. The following shall apply as test conditions of C.2.1.

C.2.1.5.1 Reagents.

a. Purified water of 18 M-ohm or greater resistivity.

b. Multi-element standards certified commercially obtained solutions.

c. Nitric Acid, concentrated, suitable for trace metal analysis.

C.2.1.5.2 Equipment.

a. Hot plate – controllable heat.

b. Inductively coupled argon plasma atomic emission spectrometer (ICP-AES). Other detectors may be used, following C.2.1 guideline.

c. Graduated PP ICP tubes/digestion cups (Environmental Express, Item No. UC475-GN or equivalent).

d. ISO 1042 certified Class A polymethylpentene (PMP) or perfluoroalkoxy- copolymer (also referred to as perfluoxy copolymer (PFA)) volumetric flasks.

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APPENDIX D

NON-VOLATILE RESIDUE

D.1 SCOPE

D.1.1 Scope. This Appendix describes three methods for the determination of non-volatile residue of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

D.2 METHODS

D.2.1 Hot plate method. This method involves the gradual evaporation of a hydrazine sample on an explosion proof hot plate in a fume hood. The difference in weight before and after evaporation is calculated as NVR.

D.2.1.1 Procedure. Use one of the following procedures. D.2.1.1.1 Method when using water that has been certified as containing no NVR. D.2.1.1.2 Method when the water's NVR content is unknown or not certified.

D.2.1.1.1 Method for certified NVR water. Place two platinum evaporating dishes in an oven at 110 °C for 30 min., labelled as samples A1 and A2. Remove evaporating dishes from the oven, place in a desiccator to cool for 15 min, and then place next to a five place balance long enough to equilibrate (usually around 30 min). Weigh each clean, dry evaporating dish to the nearest 0.0001 g until the weight remains constant to ± 0.1 mg. Record the sample evaporating dish weight as A1 and A2. Add 40 mL of water (ASTM D1193 Type II or better) to each one of the evaporating dishes. Water using this method must be pre-certified as having no NVR, either due to filtering and cleaning by the testing laboratory or certified as no NVR from the manufacturer or supplier. Transfer 40 mL of hydrazine sample into the evaporating dishes marked A1 and A2. Hydrazine and water form an azeotrope (constant boiling mixture) with a maximum boiling point of 120.5 °C (771 mmHg pressure). Therefore, when anhydrous hydrazine and water are mixed, a corresponding eutectic of 76.0 mole percent hydrazine is the maximum concentration obtained. This eliminates any significant hazard in evaporating a hydrated sample of hydrazine to dryness. Place each evaporating dish on a hot plate which is set to a temperature of 150 – 200 °C. Evaporate to dryness, but do not bake. Remove the evaporating dishes from the explosion proof hot plate and place them in oven set at 110 °C for 30 min. Remove dishes from the oven and place in a desiccator for 15 min to cool; then place them next to a five place balance long enough to equilibrate. Weigh each evaporating dish as before to a constant weight. (NOTE: A/C fluctuations/cycling may significantly affect the weights obtained for the evaporating dishes; a narrow range viscosity thermometer is recommended for monitoring room temperature during the weighing process). Record the sample evaporating dish weight as B1 and B2.

D.2.1.1.2 Method for non-certified water. Place three platinum evaporating dishes in an oven at 110 °C for 30 min. Label one as a blank and the others as samples A1 and A2. Remove evaporating dishes from the oven, place in a desiccator to cool for 15 min, and then place next to a five place balance long enough to equilibrate (usually around 30 min). Weigh each clean, dry evaporating dish to the nearest 0.0001 g until the weight remains constant to ± 0.1 mg. Record the sample evaporating dish weight as A1, A2 and the blank evaporating dish weight as C. Add 40 mL of water (ASTM D1193 Type II or better) to each one of the evaporating dishes. Transfer 40 mL of hydrazine sample into the evaporating dishes marked A1 and A2. Hydrazine and water form an azeotrope (constant boiling mixture) with a maximum boiling point of 120.5 °C (771 mmHg pressure). Therefore, when anhydrous hydrazine and water are mixed, a corresponding eutectic of 76.0 mole percent hydrazine is the maximum concentration obtained. This eliminates any significant hazard in evaporating a hydrated sample of hydrazine to dryness. Place each evaporating dish on a hot plate which is set to a temperature of 150 – 200 °C. Evaporate to dryness, but do not bake. Remove the evaporating dishes from the explosion proof hot plate and place them in oven set at 110 °C for 30 min. Remove dishes from the oven and place in a desiccator for 15 min to cool; then place them next to a five place balance long enough to equilibrate. Weigh each evaporating dish as before to a constant weight. (NOTE: A/C fluctuations/cycling may significantly affect the weights obtained for the evaporating dishes; a narrow range viscosity thermometer

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is recommended for monitoring room temperature during the weighing process). Record the sample evaporating dish weight as B1, B2 and the blank evaporating weight as D.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) and elemental analysis (4.3.5) determinations.

D.2.1.2 Calculation. Calculate the non-volatile residue of the hydrazine as follows for each dish. Report the average of the two NVR values:

$$NVR\ mg/L = \left[\frac{(Wt_B - Wt_A)}{Sample\ volume_{mL}}\right] \times 10^6 \quad \text{(for procedure D.2.1.1.1)}$$

Or

$$NVR\ mg/L = \left[\frac{(Wt_B - Wt_A) - (Wt_D - Wt_C)}{Sample\ volume_{mL}}\right] \times 10^6 \text{ (for procedure D.2.1.1.2)}$$

where,

A = Weight of the sample evaporating dish, g.

B = Weight of the sample evaporating dish and residue, g.

C = Weight of blank evaporating dish, g.

D = Weight of blank and water residue, g.

D.2.1.3 Equipment.

a. Hot plate, explosion proof.

b. High efficiency fume hood.

c. Five decimal place analytical balance (semi-micro balance).

D.2.2 Dropwise method.

D.2.2.1 Procedure. Clean the unassembled apparatus with detergent and rinse thoroughly with purified water. Dry the parts in an oven at 110 °C. Assemble the apparatus as shown in Figure D-1. Fill the water bath with purified water and turn on the heater. Adjust the water level of the bath to approximately one-half inch below the lowest fitting and mark this level on the bath. When the temperature bath reaches boiling, adjust the N_2 purge to ~ 2 L/min and turn on the aspirator. With the sample delivery valve closed, the pressure in the apparatus should be below 30 mmHg. Open the sample delivery valve slowly to adjust the apparatus pressure to ~ 60 mmHg. Allow the apparatus to purge for ~ 5 min to remove traces of water. Remove the water bath and dry the outside of the residue vessel with a clean lintless tissue. Disconnect the aspirator from the apparatus and remove the residue vessel using a tissue to hold the vessel. Wipe the outside of the vessel and place it near the analytical balance. After at least 20 min, accurately weigh the residue vessel until the weight remains constant to ± 0.1 mg. Record the weight. Reassemble the apparatus and close the sample delivery valve. Replace

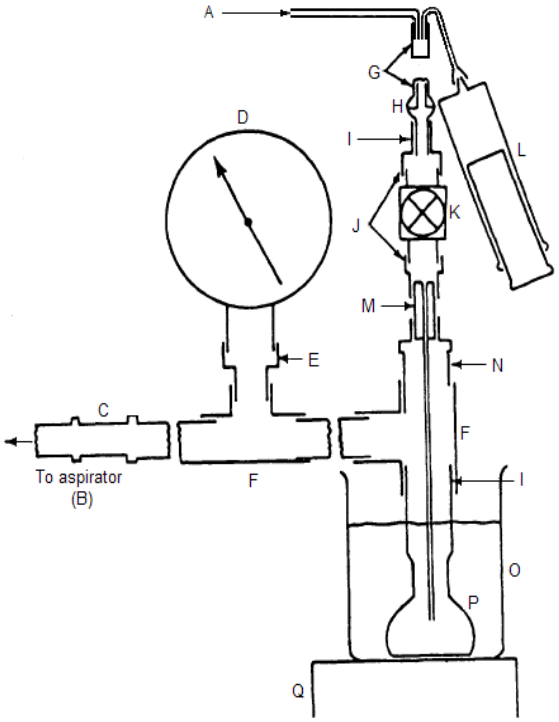
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the water bath, adjust the water level to the mark, and heat the water to boiling. Transfer 100 mL of hydrazine to the apparatus. With the N_2 purge in place, turn on the water aspirator and reduce the apparatus pressure to less than 30 mmHg. Slowly open the sample delivery valve to admit hydrazine to the residue vessel and adjust the flow such that the apparatus pressure does not exceed 60 mmHg (~ 2 drops per second). The quantity of hydrazine in the residue vessel should not exceed 2 - 4 mL at the equilibrium evaporation rate. Adjust the sample delivery valve to maintain the evaporation pressure at less than 60 mmHg. After 100 mL of hydrazine have evaporated (~ 30 min), allow the residue vessel to purge for an additional 5 min to ensure complete evaporation of volatile material. Remove the water bath, disconnect the aspirator, wipe the outside of the residue vessel, and remove it from the apparatus. Allow the residue vessel to stand near the analytical balance for at least 20 min before weighing. Weigh the residue vessel as above and record the weight. To clean the residue vessel reflux 5 mL of 5N ACS reagent grade H_2SO_4 (or equivalent) in the vessel for several minutes followed by several rinses of purified water. The vessel may be dried as described above or by gently heating the vessel with a burner while purging with dry N_2. Regardless of the drying method used, allow at least 20 min for the vessel to equilibrate to ambient temperature and humidity before weighing.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) determination.

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A - GN_2

B - Water aspirator (glass or polyethylene)

C - Connector (polyethylene, separable)

D - Gauge (stainless steel, absolute pressure, 0 - 760 mmHg range)

E - Adapter (Swagelok SS-8-TA-7-4 or equivalent)

F - Union Tee (Swagelok SS-810-3 or equivalent; 2 ea.)

G - Pipette (100 mL, volumetric, borosilicate glass)

H - Joints (borosilicate glass, spherical, 12/5, ball, socket and clamp)

I - PTFE Ferrules (Swagelok T-814-1, T-813-1, T-404-1, T-403-1, or equivalent)

J - Connector (Swagelok SS-400-7-2 or equivalent)

K - Sample delivery valve (Swagelok SS-ORM2 or equivalent)

L - Syringe (100 mL glass with an 8 – 12 in PTFE needle).

M - Rod (PTFE or polyethylene, 1/4 in OD & a 1/16 in OD x 0.025 in wall tube inserted)

N - Reducer (Swagelok SS-400-R-8 or equivalent)

O - Beaker (glass, 1000 mL capacity)

P - Flask (25 mL volumetric, borosilicate glass)

Q - Hot plate (adjustable heat range)

NOTE: All fittings, stainless steel, Swagelok or equivalent.

FIGURE D-1. *NVR apparatus.*

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D.2.2.2 Calculation. Calculate the non-volatile residue of the hydrazine as follows:

$$NVR\ mg/L = \left[\frac{W_2 - W_1}{V}\right] \times 10^6$$

where,

W_1 = Weight of clean vessel, g.

W_2 = Weight of residue and vessel, g.

V = Volume of sample used, mL.

Report the averaged result of duplicate NVR determinations.

D.2.2.3 Equipment. The following equipment shall apply as test conditions of D.2.2.

a. Components described in Figure D-1.

b. Tubing – polyethylene, ½ in OD, borosilicate glass, ½ in OD.

c. Clamps – hose, to fit ½ in OD tube.

d. Five decimal place analytical balance (semi-micro balance).

D.2.3 Rotary evaporation.

D.2.3.1 Procedure. Dry a 250 mL round bottom flask in an oven at 110 °C for one hour. Allow the flask to cool in a desiccator and weigh on the analytical balance until the weight remains constant to ± 0.1 mg. Transfer 100 mL of hydrazine into the 250 mL flask. Place the flask on the evaporator using a standard taper Teflon sleeve as a seal. Attach a water aspirator pump to the evaporator via a one liter suction flask and evaporate the hydrazine to dryness. A water bath may be placed around the rotating flask to aid evaporation.

WARNING

The contact of hydrazine vapor with metal surfaces, as in a metal rotary evaporator, must be avoided. Do not exceed 40 °C during the initial evaporation phase when hydrazine liquid is present. Chemical splash proof goggles must be worn throughout the entire evaporation process.

When no more hydrazine evaporates, the water bath is lowered. Final high boiling components (mainly aniline) are distilled under vacuum at temperatures up to 100 °C using a hot air gun for heating. The flask is then removed and dried in an oven at 110 °C for one hour and weighed as before to a constant weight.

NOTE

Save the propellant residue/hydrazine NVR for chloride (4.3.4) determination.

D.2.3.2 Calculation. Use equation in D.2.2.2.

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D.2.3.3 Equipment.

a. All glass rotary evaporator, BÜCHI Rotavapor or equivalent.

b. Flask, round bottom, 250 mL, with standard neck to fit evaporator.

c. Water aspirator – glass or polyethylene.

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CARBON DIOXIDE

E.1 SCOPE

E.1.1 Scope. This Appendix describes the method for the determination of carbon dioxide content of the hydrazine. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance. Determination of the CO_2 content in the hydrazine shall be performed before all other analyses to prevent further absorption of CO_2.

E.2 METHOD

E.2.1 Determination of carbon dioxide by non-dispersive IR analysis. This method is to be utilized to determine the CO_2 concentration in hydrazine within a 10 - 100 mg/kg (ppmw) range. The sample of hydrazine is injected into a solution of 30% phosphoric acid which absorbs the hydrazine and ammonia. Carbon dioxide is liberated from the acidic solution and is swept out of the solution by a stream of N_2 gas through a silica gel drying tube to remove water vapor and is measured by an infrared technique.

E.2.1.1 Calibration procedure.

a. Assemble the apparatus as shown in Figure E-1.

b. Adjust the flow of N_2 purge gas according to the manufacturer's recommendation by use of the control valve and flow meter.

c. Add an appropriate amount of 30% phosphoric acid containing several drops of bromocresol green or methyl red indicator to the CO_2 reaction flask, turn on the N_2 gas to give vigorous, constant mixing. The N_2 gas flow (~ 200 mL/min) may need adjustment to obtain reproducible results but must remain constant throughout the analysis.

d. Allow the instrument to stabilize and adjust the zero.

E.2.1.2 Standardization.

If using direct instrument response comparison (flow technique),

a. Inject via the septum on the phosphoric acid containing unit, an appropriate amount of a prepared 100 ng/µL (ppmv) working CO_2 carbonate or bicarbonate standard into the phosphoric acid solution to achieve a reading of 100 ppm on the instrument (~ 200 mL). Repeat the injection to ensure repeatability is within ± 10 ppm and record.

b. Once repeatability is verified, repeat as above with ~ 100 µL (50 ppm) and 50 µL (25 ppm) and check for linearity.

If using a data acquisition device,

a. Prepare a 5 point calibration curve using the 100 ppmv working standard to bracket the expected concentration range of the samples using a gas-tight syringe (10 µL is equivalent to 1000 ng).

b. Plot the absorbance counts of the standards vs. nanograms. to determine the CO_2 concentration in the samples. The calibration correlation coefficient shall be ≥ 0.995.

E.2.1.3 Sample analysis. Inject an appropriate amount of hydrazine sample into the phosphoric acid reaction flask to obtain a reproducible result within the standards range. Record the reading from the

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infrared analyzer scale or data acquisition device (from the zero point). Perform the sample analysis in triplicates (verifying reproducibility) and use the average to determine the CO_2 concentration.

E.2.1.3.1 Calculations. Determine the carbon dioxide content of the hydrazine as follows,

a. Flow technique,

$$CO_2\ Wt\% = \frac{(Std\ conc_{ppmv})(Std\ vol_{\mu L})(Sample\ response)}{(Sample\ vol_{\mu L})(Std\ response)(\rho)(1 \times 10^4)}$$

where,

ρ = Density of hydrazine (1.01 g/mL).

b. Data acquisition device,

$$CO_2\ Wt\% = \frac{Sample\ CO_{2\ ng}}{(Sample\ vol_{\mu L})\ (\rho)(1 \times 10^4)}$$

where,

ρ = Density of hydrazine (1.01 g/mL).

E.2.1.4 Reagents.

a. Purified water, CO_2 free - Prepare by boiling purified water vigorously for 15 min or by bubbling N_2 gas through it for several hours.

b. Carbon Dioxide Stock Solution, 1000 ppm - Dissolve 2.4083 g ACS reagent grade sodium carbonate or 1.9088 g ACS reagent grade sodium bicarbonate, previously dried for one hour at 100 °C, in CO_2 free purified water in a 1 L volumetric flask; fill to the mark and mix thoroughly. Transfer 1 mL of this solution to a 10 mL volumetric flask, fill to the mark with CO_2 free purified water to prepare a 100 ppm standard. Prepare fresh daily.

c. Phosphoric Acid, 30% by volume - Dilute 355 mL of 85% H_3PO_4 to 1 L with purified water. Add enough indicator to this solution to readily determine when the acid solution is depleted.

d. Dry N_2 gas supply.

e. Silica gel desiccant.

E.2.1.5 Equipment.

a. Non-dispersive infrared analyzer, Teledyne GFC-7000T or equivalent.

b. Glass reaction tube with replaceable septum.

c. PTFE tubing.

d. Glass drying tube.

e. Needle valve to control flow into reaction tube.

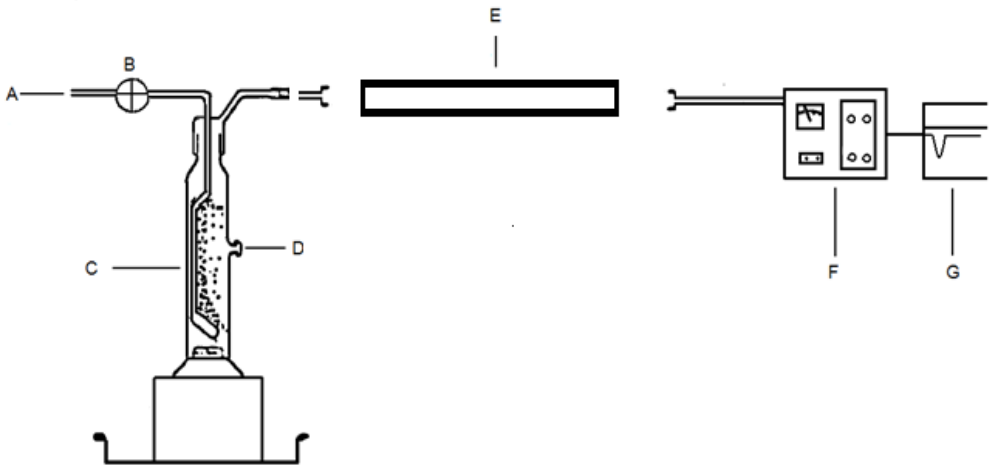
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f. Fittings to achieve airtight connections.

g. Operating fume hood.

h. Gas-tight syringe.

i. Suitable data acquisition device.



A	-	Sparge gas inlet	E	-	Silica gel drying tube for removing H_2O
B	-	Needle valve	F	-	Infrared analyzer
C	-	Reactor and phosphoric acid solution	G	-	Recorder
D	-	Septum inlet for hydrazine			

FIGURE E-1. *CO_2 apparatus.*

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CARBONACEOUS MATERIAL, TOLUENE, AND ANILINE

F.1 SCOPE

F.1.1 Scope. This Appendix outlines the mandatory method for determining carbonaceous material content in hydrazine. Additionally, it details the procedure for measuring aniline and, if necessary, toluene content. Compliance with the information presented herein is required for specification adherence.

F.2 METHOD

F.2.1 Gas chromatographic method.

F.2.1.1 Procedure. Use a 30-meter (m) × 0.53-millimeter (mm), 1.00 µm film thickness, wax-phase (DB-Wax, Stabilwax DB, or equivalent) column, an FID equipped with a split injector, and an ultra-inert, deactivated, glass wool packed split liner shall be used. Table F-I provides a suggested list of the GC parameters for initial setup. These parameters may vary depending on GC system and detector and may be adjusted as necessary to provide adequate component resolution.

TABLE F-I. *Recommended GC parameters.*

PARAMETER	CONDITION[1]
Injector Port Temp	175 °C
Injection Mode	Split injection
Sample Volume.	1 µL
Split Ratio	2:1
Carrier Gas	He or H_2
Control Flow	10 mL/min
Control Mode	Flow
FID Detector Temp.	210 °C
Equilibration Time	2.00 min
Initial Temp.(T_0)	60 °C
Initial Time (t_0)	3.00 min
Rate (1)	10 °C/min
Final Temp. (T_{f1})	100 °C
Final Time (t_{f1})	0.00 min
Rate (2)	20 °C/min
Final Temp. (T_{f2})	185 °C
Final Time (t_{f2})	3.75 min
Note 1 – Parameters identified are for a He carrier. H_2 carrier may require adjustments.	

The elution order is as follows:

UDMH < IPA < MMH < toluene < hydrazine < aniline

Figure F-1 shows a typical chromatogram illustrating elution order.

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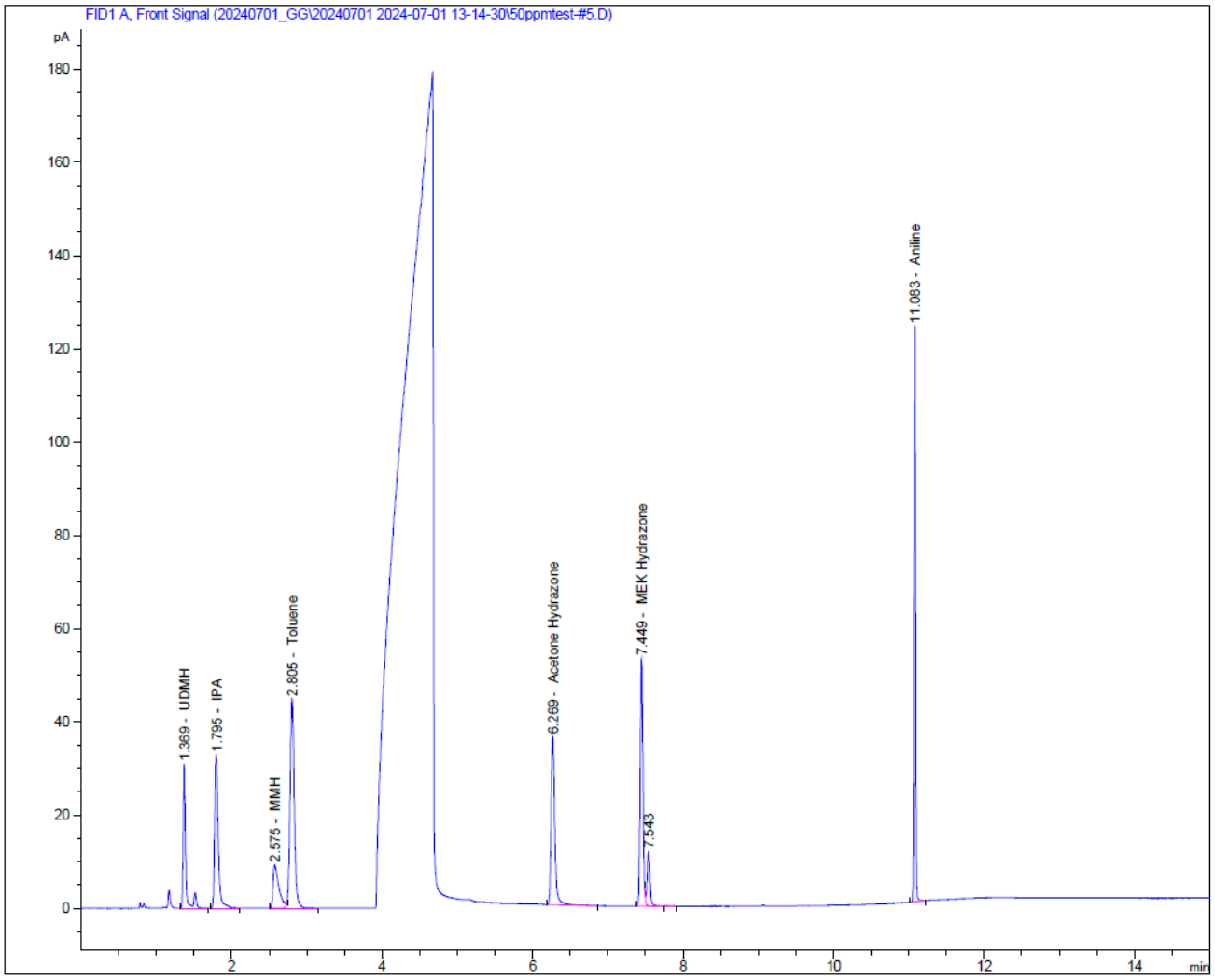


FIGURE F-1. *Typical chromatogram.*

F.2.1.2 Calibration. Prepare a standard stock solution, and a working standard of IPA, MMH, aniline, and toluene (if required) in hydrazine (N_2H_4) as described in F.2.1.2.1 and F.2.1.2.2 (deviations are allowed as long as the working standard is conducive to the analysis). UDMH can be calibrated using the MMH response, it is not necessary in the working standard. It is suggested to run a UDMH standard to verify retention time. Run a blank of high purity N_2H_4 meeting this specification and subtract the areas of UDMH, IPA, MMH, and toluene from the areas of the standard. Determine the area of all impurities providing a positive FID response with a signal to noise ratio of at least 3:1 after the injection peak. The sensitivity of the system shall be such that it allows for the effective quantitation of at least 5 ppmw MMH and at least 1 ppm aniline. Analysis is to be conducted in triplicate, with the RSD of the total carbonaceous area less than 10%.

F.2.1.2.1 Standard stock solution (5000 ppmw for IPA, MMH, and toluene; 1000 ppmw aniline).

1. Weigh a 40 mL volatile organic analysis (VOA) type septa vial with its cap to the nearest 0.0001 g and record the weight.

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2. Add ~ 30 mL of high purity N_2H_4 meeting this specification to the vial; cap, weigh to the nearest 0.0001 g, and record the weight. The difference in weight before and after the addition is the amount of N_2H_4 added.

3. Using a gas-tight glass syringe add ~ 190 µL IPA to the vial, injected through the septum, and weigh to the nearest 0.0001 g and record the weight. Subtract the weight of the high purity N_2H_4 from the weight obtained to determine the weight of IPA added.

4. Follow same procedure as above with the additions of ~175 µL MMH, ~30 µL aniline, and ~175 µL toluene (if required). For analysis of MPH grade, add ~490 µL aniline (~0.5% weight).

5. Mix the solution thoroughly and store tightly sealed in the fuel hood under a N_2 gas purge.

6. An alternate volume or concentration of the standard stock solution may be prepared depending on laboratory needs, provided the volumes of analytes added are within a suitable tolerance to accurately obtain the mass of the amount added.

F.2.1.2.2 Working standard.

1. Weigh a suitable size VOA-type vial with its cap to the nearest 0.0001 g. Record the weight.

2. Add an appropriate amount of high purity N_2H_4 to the vial, cap, and weigh following the same procedure as in F.2.1.2.1. The difference in weight before and after the addition is the amount of N_2H_4 added.

3. Shake the standard stock solution and inject into the vial to obtain a mid-range concentration based on the maximum allowed limits for high purity and monopropellant N_2H_4 and weigh to the nearest 0.0001 g. Recommend a 1:100 dilution for ~ 50 ppmw for all components of interest except aniline at ~10 ppmw. Subtract the weight of the high purity N_2H_4 from the weight obtained to determine the weight of standard stock solution added.

F.2.1.3 Calculations.

a. Standard stock solution:

$$Analyte\ ppmw_{Stock\ Std} = \frac{Wt_{Analyte} \times 10^6}{Wt_{Total}}$$

where,

$$Wt_{Total} = Wt_{N_2H_4} + Wt_{IPA} + Wt_{MMH} + Wt_{aniline} + Wt_{toluene}$$

b. Working standard:

$$Analyte\ ppmw_{Wk\ Std} = \frac{Analyte\ ppmw_{Stock\ Std} \times Wt_{\ Stock\ Std\ added}}{Wt_{Stock\ Std\ added} + Wt_{N_2H_4}}$$

c. Sample calculation:

$$Analyte\ ppmw_{Sample} = \frac{Analyte\ ppmw_{Wk\ Std} \times A_s}{A_{analyte}}$$

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where,

$A_{analyte}$ = The area of the analyte peak. MMH can be used for the calculation of UDMH.

Calculate non-calibrated carbonaceous material in weight percent by addition of all impurities providing a positive FID response ahead of and trailing the hydrazine peak. Calculate the content in the N_2H_4 sample based on the MMH factor as follows,

$$non - calibrated\ carbonaceous\ material\ wt\% = \left[\frac{MMH\ ppmw_{Wk\ Std} \times A_i}{A_{MMH}}\right] \times 10^{-4}$$

where,

A_i = The average (based on triplicates) of the sum of all the peak areas excluding calibrated components and aniline.

A_{MMH} = The average peak area obtained for the MMH standard.

Calculate carbonaceous material in weight percent using the following equation,

Carbonaceous material (excluding aniline) wt% = UDMH wt% + IPA wt% + MMH wt% +Toluene wt% (at point of manufacture only) + non-calibrated carbonaceous materials wt%.

At point of manufacture, report the toluene as included with the carbonaceous material as well as separately as calculated. Aniline shall be reported separately from carbonaceous material and is not included in that calculation.

F.2.1.4 Reagents and equipment. The following reagents and equipment shall apply as test conditions of F.2.1.

F.2.1.4.1 Reagents.

a. N_2H_4 - High purity grade conforming to this specification.

b. IPA - ACS reagent grade.

c. MMH - Conforming to MIL-PRF-27404.

d. Aniline - ACS reagent grade.

e. Toluene - ACS reagent grade.

f. UDMH - Conforming to MIL-PRF-25604 (if being used).

F.2.1.4.2 Equipment.

a. Gas chromatograph equipped with an FID and a split injector.

b. Agilent ChemStation or equivalent.

c. A 30 m x 0.53 mm ID and 1.00 µm film thickness Stabilwax®-DB column or equivalent.

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d. Ultra-inert deactivated liner, split, straight, glass wool.

e. No-bleed septa.

f. Gas-tight syringe.

g. Analytical balance calibrated to 0.0001 g.

F.2.1.5 Quality control. The following performance criteria must be met for carbonaceous material analysis:

GC replicate analyses (3 consecutive) on total carbonaceous peak area	RSD[1] ≤ 10 %

1. RSD = Standard Deviation * 100% / Mean

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CONCLUDING MATERIAL

Custodians:
Navy – AS
Air Force – 68
DLA – PS

Preparing activity:
Air Force – 68
(Project: 9135-2024-008)

Review activities:
Air Force - 19

Civil Agency
NASA - NA

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